## 3.2. METHODS FOR SOLVING D.C. CIRCUIT

**3.2.1. Maxwell loop (Mesh) method.** This method is based on Kirchhoff's voltage law. Let us consider the circuit shown in Fig. 3.2. Now select the loops as per your choice and circulate the current in each loop in any direction. Assure that all the elements (active and passive) must have at least one current through them.

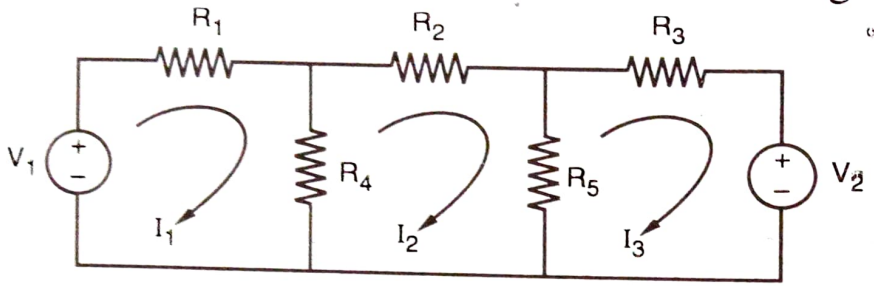



Fig. 3.2.

Now apply KVL in each loop :

In loop 1 : $$V_1 = R_1 I_1 + R_4 (I_1 - I_2) \qquad ...(3.1)$$

In loop 2 : $$0 = R_4 (I_2 - I_1) + R_2 I_2 + R_5 (I_2 - I_3) \quad \text{...(3.2)}$$

In loop 3 : $$-V_2 = R_5 (I_3 - I_2) + R_3 I_3 \quad \text{...(3.3)}$$

by solving these three equations, we can calculate the value of currents $I_1$, $I_2$ and $I_3$.

Alternatively the value of currents can also be determined by using the Cramer's rule (Matrix method). For this we have to arrange the equations 3.1, 3.2 and 3.3 in following way :

$$(R_1 + R_4) I_1 - R_4 I_2 + 0 I_3 = V_1 \quad \text{...(3.4)}$$

$$-R_4 I_1 + (R_2 + R_4 + R_5) I_2 - R_5 I_3 = 0 \quad \text{...(3.5)}$$

$$0I_1 - R_5 I_2 + (R_3 + R_5) I_3 = -V_2 \quad \text{...(3.6)}$$

Now write down the equations. 3.4, 3.5 and 3.6 in matrix form

$$\begin{bmatrix} R_1 + R_4 & -R_4 & 0 \\ -R_4 & R_2 + R_4 + R_5 & -R_5 \\ 0 & -R_5 & R_3 + R_5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ 0 \\ -V_2 \end{bmatrix} \quad \text{...(3.7)}$$

or $$[R]\ [I] = [V] \quad \text{...(3.8)}$$

where $[R]$ = Resistance matrix, $[I]$ = Current matrix, $[V]$ = Voltage matrix

The resistance matrix is symmetrical about the diagonal elements and equation 3.7 is in the form of Ohms law ($RI = V$).

Let $\Delta_R$ is the determinant of resistance matrix then all three currents $I_1$, $I_2$ and $I_3$ can be determined by Cramer's rule.

For finding the current is loop 1, we apply the Cramer's rule : According to which the current $I_1$ will be ratio of two determinants. In which denominator is the determinant of resistance matrix $\Delta_R$ and in numerator, first column of resistance matrix is replaced by the elements of voltage matrix

*i.e.*,
$$I_1 = \frac{\begin{vmatrix} V_1 & -R_4 & 0 \\ 0 & R_2 + R_4 + R_5 & -R_5 \\ -V_2 & -R_5 & R_3 + R_5 \end{vmatrix}}{\Delta_R} \quad \ldots(3.9)$$

Similarly
$$I_2 = \frac{\begin{vmatrix} R_1 + R_4 & V_1 & 0 \\ -R_4 & 0 & -R_5 \\ 0 & -V_2 & R_3 + R_5 \end{vmatrix}}{\Delta_R} \quad \ldots(3.10)$$

and
$$I_3 = \frac{\begin{vmatrix} R_1 + R_4 & -R_4 & V_1 \\ -R_4 & R_2 + R_4 R_5 & 0 \\ 0 & -R_5 & -V_2 \end{vmatrix}}{\Delta_R} \quad \ldots(3.11)$$

**3.2.2. Maxwell Node method.** Again consider the Fig. 3.2, which is redrawn in Fig. 3.3.

For applying Maxwell node method, we select nodes. In network of Fig. 3.3, we select three nodes, node *a*, node *b* and one reference (ground) node. Since voltage is not the absolute quantity and is always defined with respect to some reference or ground so we have to choose one node as the reference node or ground (Datum node) whose voltage is zero. Now take variable $v_a$ as the voltage for node '*a*' and $v_b$ for node '*b*'.

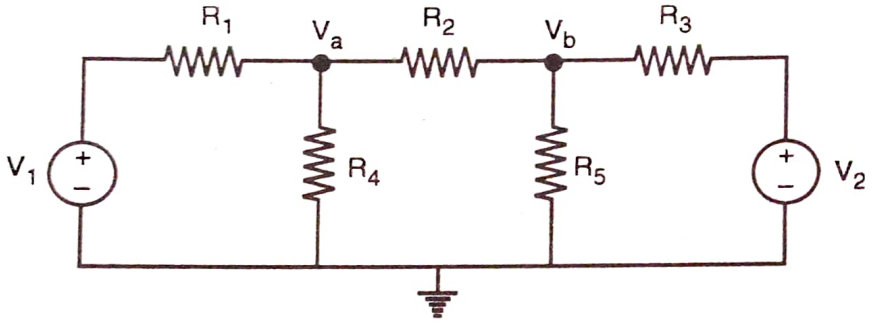
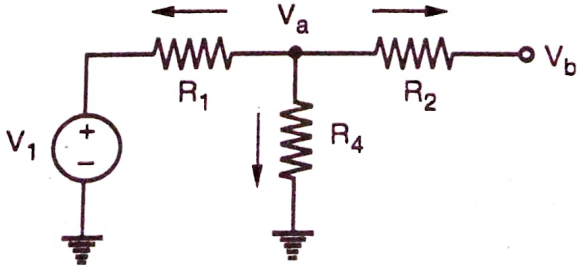


Fig. 3.3. Maxwell Node method.

As in Maxwell loop method we apply Kirchhoff's voltage law in each loop, in Maxwell node method we apply Kirchhoff's current law (KCL) at each node.

According to KCL, algebraic summation of currents meeting at a node is equal to zero.

**KCL at node '*a*'.** Assume that all the currents leaving the node '*a*'. Redraw the circuit of Fig. 3.3 for node '*a*' only. (Fig. 3.4).




By KCL at node '*a*'

$$\frac{v_a - V_1}{R_1} + \frac{v_a}{R_4} + \frac{v_a - v_b}{R_2} = 0 \qquad ...(3.12)$$

Arranging the above equation,

$$\left[\frac{1}{R_1}+\frac{1}{R_4}+\frac{1}{R_2}\right]v_a-\frac{1}{R_2}v_b=\frac{V_1}{R_1} \qquad ...(3.13)$$

Fig. 3.4.

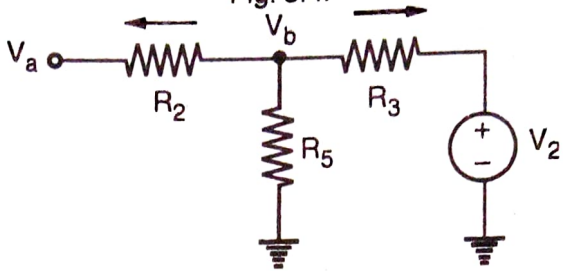



Fig. 3.5.

Now draw the circuit of Fig. 3.3 for node '*b*' only. (Fig. 3.5).

By KCL at node '*b*'

$$\frac{v_b-v_a}{R_2}+\frac{v_b}{R_5}+\frac{v_b-V_2}{R_3}=0 \qquad ...(3.14)$$

Arranging this equation,

$$-\frac{1}{R_2}v_a+\left[\frac{1}{R_2}+\frac{1}{R_5}+\frac{1}{R_3}\right]v_b=\frac{V_2}{R_3} \qquad ...(3.15)$$

By solving equations 3.13 and 3.15 we can determine the values of $v_a$ and $v_b$.

Alternatively the values of $v_a$ and $v_b$ can also be determined by Matrix or Cramer's rule. equations 3.13 and 3.15 can be written in the form of matrix as,

$$\begin{bmatrix} \frac{1}{R_1}+\frac{1}{R_4}+\frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2}+\frac{1}{R_5}+\frac{1}{R_3} \end{bmatrix}\begin{bmatrix} v_a \\ v_b \end{bmatrix}=\begin{bmatrix} \frac{V_1}{R_1} \\ \frac{V_2}{R_3} \end{bmatrix} \qquad ...(3.16)$$

or $$[G][v] = [I]$$

where $[G]$ = Conductance $\left(\frac{1}{R}\right)$ matrix

$[v]$ = Voltage matrix

$[I]$ = Current matrix

Let $\Delta_G$ determinant of conductance matrix then by Cramer's rule,

$$v_a = \frac{\begin{vmatrix} \frac{V_1}{R_1} & -\frac{1}{R_2} \\ \frac{V_2}{R_3} & \frac{1}{R_2}+\frac{1}{R_5}+\frac{1}{R_3} \end{vmatrix}}{\Delta_G} \qquad ...(3.17)$$

and

$$v_b = \frac{\begin{vmatrix} \frac{1}{R_1}+\frac{1}{R_4}+\frac{1}{R_2} & \frac{V_1}{R_1} \\ -\frac{1}{R_2} & \frac{V_2}{R_3} \end{vmatrix}}{\Delta_G} \qquad ...(3.18)$$

## EXAMPLES ON FUNDAMENTAL LAW :

**Example 3.1.** *Find the current through 10 Ω resistor and power dissipated in the (Fig. 3.24)*

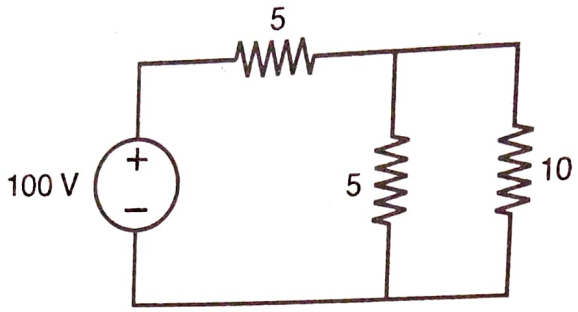



Fig. 3.24.

**Solution.** 5 and 10 Ω resistors are in parallel $= \dfrac{5 \times 10}{5 + 10} = 3.33\ \Omega$

So the circuit is redrawn as shown in Fig. 3.25.

$i$ = Current drawn from the source $= \dfrac{100}{8.33} = 12$ Amp.

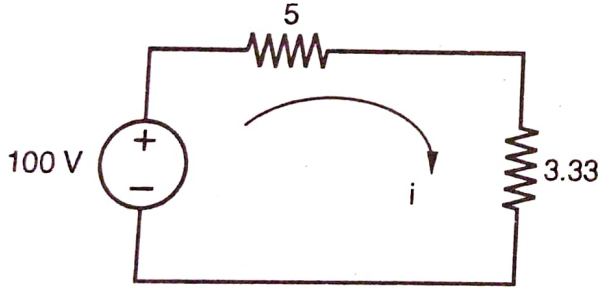



Fig. 3.25.

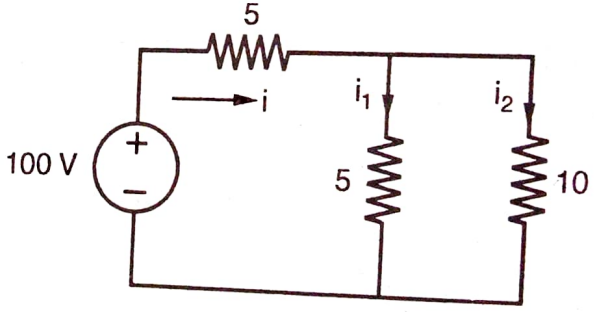



Fig. 3.26.

So the current through 10 Ω resistor can be determined by current division rule,

$$i_2 = i \times \frac{5}{5 + 10} = 4 \text{ Amp.}$$

So the power dissipated in 10 Ω resistor, $P = [i_2]^2 \times 10 = 160$ Watts.

**Example 3.2.** *In the given network, find the value of current source, if the voltage drop across 5 Ω resistor is 10 volts.*

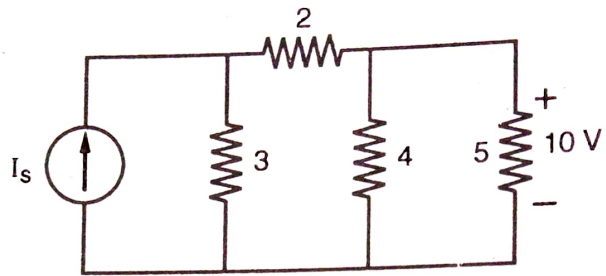



Fig. 3.27.

**Solution.** Let the current through 5 Ω is $i_1$ and that is through 4 Ω is $i_2$.

$$i_1 = 10/5 = 2 \text{ Amp and } i_2 = 10/4 = 2.5 \text{ Amp}$$

(Since voltage across 4 Ω is also 10 V, because it is in parallel to 5 Ω resistor)

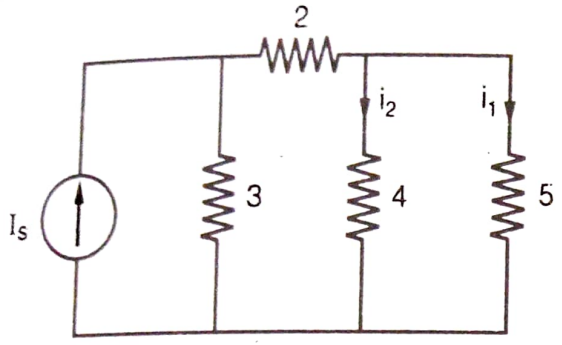



Fig. 3.28.

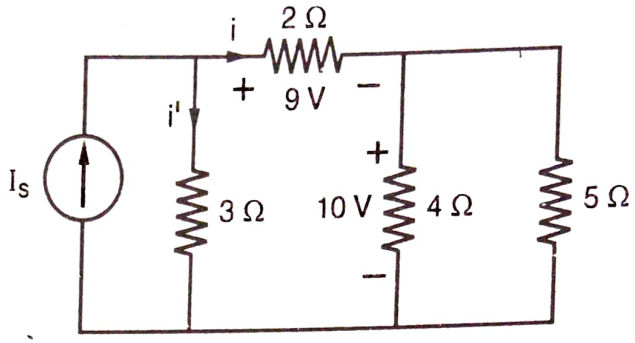



Fig. 3.29.

Now the current through 2 Ω resistor is $i = i_1 + i_2 = 4.5$ Amp

So the voltage across the 2 Ω resistor is $= 2 \times 4.5 = 9$ volts.

Voltage across the 3 Ω resistor is 9 + 10 = 19 V. So the current through 3 Ω resistor is $i' = 19/3 = 6.333$ Amp.

Now the source current is

$$I_s = i + i' = 4.5 + 6.333 = 10.83 \text{ Amp.}$$

**Example 3.3.** *Find the current through each element and potential difference across 15 Ω resistor.*

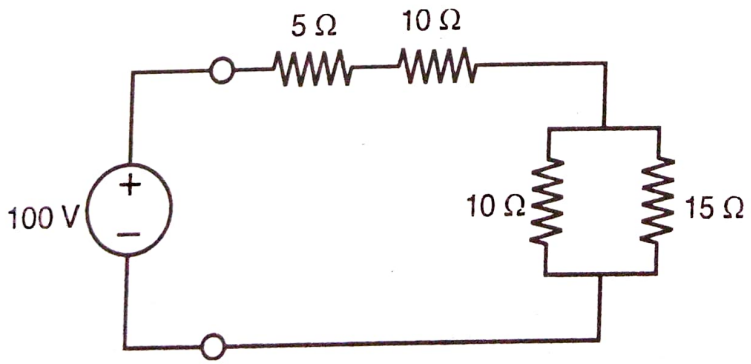



Fig. 3.30.

**Solution.**

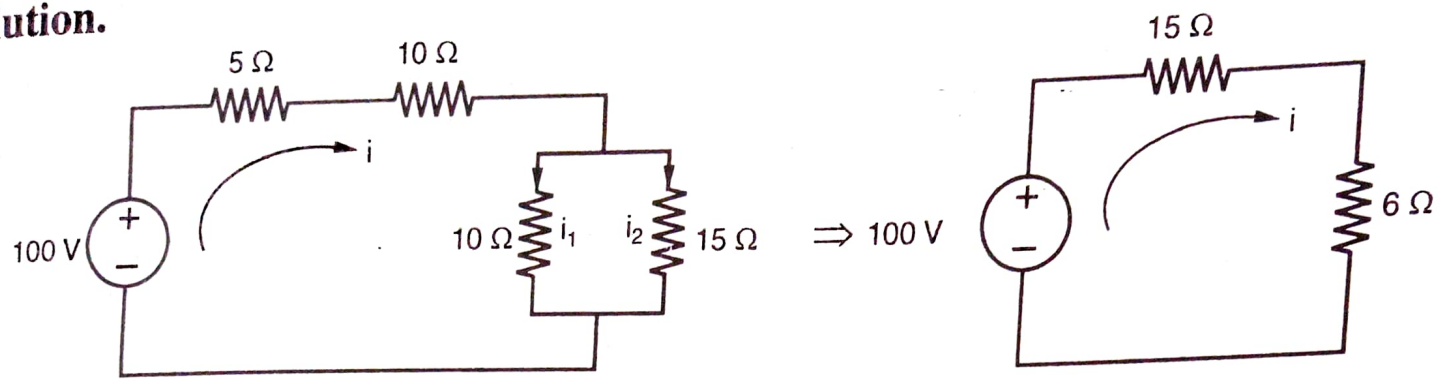



Fig. 3.31.

The total current through the circuit,

$$i = \frac{V}{R} = \frac{100}{15+6} = \frac{100}{21} = 4.76 \text{ Amp.}$$

∴ Current through 5 Ω and 10 Ω resistor will be

$$i = 4.76 \text{ Amp.}$$

Now, current through 10 Ω resistor which is in parallel, will be

$$i_1 = i \times \frac{r_2}{r_1 + r_2} \{\text{by Current division rule}\}$$

$$= 4.76 \times \frac{15}{10+15} = 4.766 \times \frac{15}{25} = 2.85 \text{ amp.}$$

and current through 15 Ω resistor will be

$$i_2 = i \times \frac{r_1}{r_1 + r_2} \qquad \{\text{by Current division rule}\}$$

$$= 4.76 \times \frac{10}{10 + 15} = 4.76 \times \frac{10}{25} = 1.9 \text{ Amp.}$$

∴ Potential difference across 15 Ω resistor will ee

$$v = i_2 \times r_2 = 1.9 \times 15 = 28.5 \text{ volts.}$$

**Example 3.4.** *Find the current through each element .*

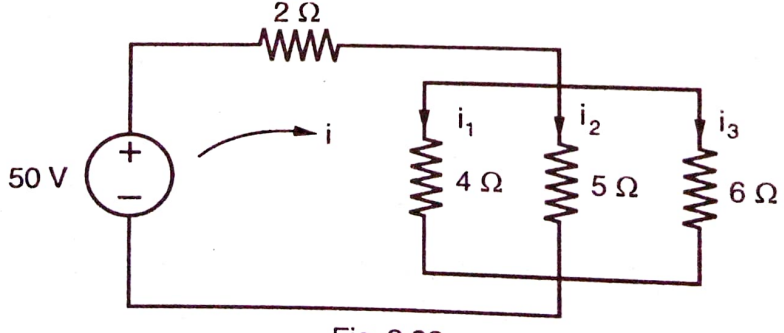



Fig. 3.32.

Fig. 3.33.

$\therefore$ The total current through the circuit will be

$$i = \frac{V}{R} = \frac{50}{2 + 1.62} = \frac{50}{3.62} = 13.81 \text{ Amp.}$$

$\therefore$ Voltage drop across parallel resistors

$$v = i \times r = 13.81 \times 1.62 = 22.37 \text{ volts.}$$

$\therefore$ The current through each element of the circuit.

(*i*) Through 2 Ω resistor, $i = 13.81$ Amp.

(*ii*) Through 4 Ω resistor, $i_1 = \frac{22.37}{4} = 5.59$ Amp.

(*iii*) Through 5 Ω resistor, $i_2 = \frac{22.37}{5} = 4.47$ Amp.

(*iv*) Through 6 Ω resistor, $i_3 = \frac{22.37}{6} = 3.73$ Amp.

**Example 3.5.** *Find the equivalent resistance across source terminals, and determine the current drawn from the source .*

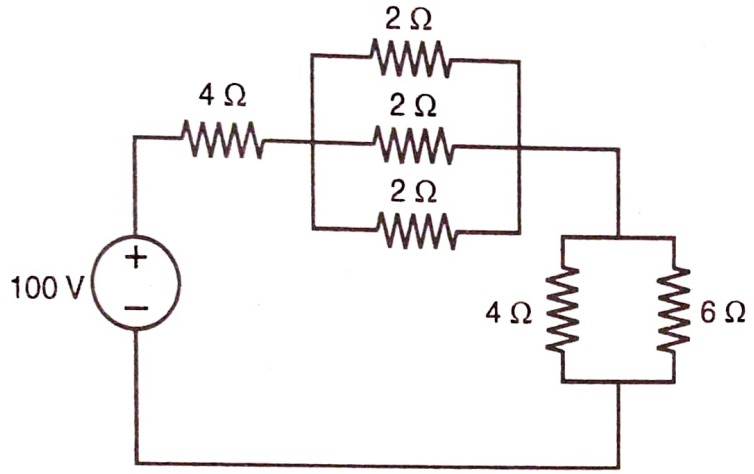



Fig. 3.34.

**Solution.** The equivalent circuit of above is

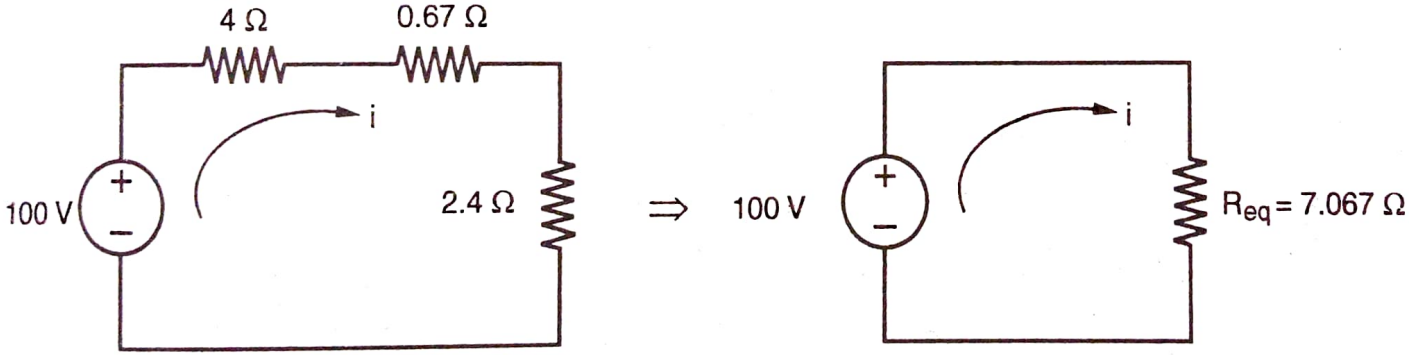



Fig. 3.35.

As all the three resistances are in series.

$$\therefore \qquad R_{eq} = 4 + 0.67 + 2.4 = 7.067\ \Omega$$

∴ The current drawn from the source,

$$i = \frac{V}{R_{eq}} = \frac{100}{7.067} = 14.16 \text{ Amp.}$$

## EXAMPLES ON MAXWELL LOOP AND NODE METHOD :

**Example 3.6.** *In the given d.c. circuit of Fig. 3.36, determine the current passing through each element.*

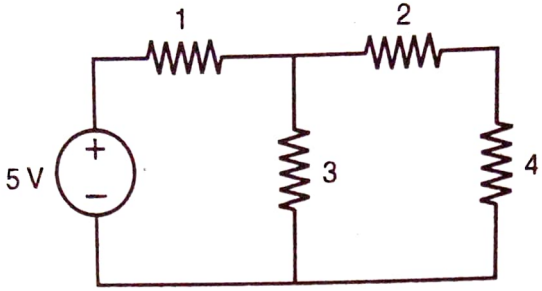



Fig. 3.36.

**Solution.** The given circuit is redrawn in Fig. 3.37 and two loop currents are made as shown in Fig. 3.37.

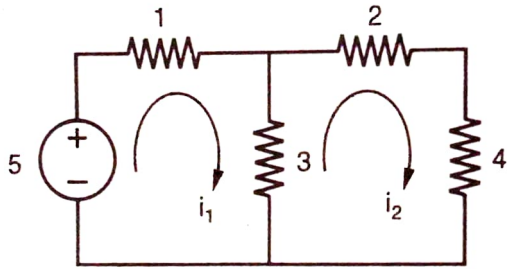



Fig. 3.37.

KVL in first loop : $1i_1 + 3\,(i_1 - i_2) = 5$ ...(*i*)

In second loop : $3\,(i_2 - i_1) + 2i_2 + 4i_2 = 0$ ...(*ii*)

Arrange equations (*i*) and (*ii*)

$$4\,i_1 - 3i_2 = 5 \quad \ldots(ii)$$

$$-3\,i_1 + 9\,i_2 = 0 \quad \ldots(iv)$$

Now we solve Equations (*iii*) and (*iv*) for current $i_1$ and $i_2$. From Equation (*iii*)

$$i_1 = \frac{3}{4} i_2 + \frac{5}{4}$$

Put this value of $i_1$ in Equation (*iv*)

$$-3\left[\frac{3}{4} i_2 + \frac{5}{4}\right] + 9\,i_2 = 0$$

$$\Rightarrow \quad -\frac{9}{4} i_2 - \frac{15}{4} + 9\,i_2 = 0$$

$$\Rightarrow \quad i_2 = 0.555 \text{ Amp.}$$

Put this value of $i_2$ in Equation (*v*), we get,

$$i_1 = \frac{3}{4}(0.555) + \frac{5}{4} = 1.667 \text{ Amp.}$$

So the current passing through 1 Ω resistor = $i_1$ = 1.667 Amp.

Current passing through 3 Ω resistor = $(i_1 - i_2)$ = 1.112 Amp.

and the current passing through 2 and 4 Ω resistor = $i_2$ = 0.555 Amp.

and the current passing through 2 and 4 Ω resistor

**Example 3.7.** *In previous example find the value of $i_1$ and $i_2$ using Cramer's rule (Matrix method).*

**Solution.** In above example, equations (*iii*) and (*iv*) can be written in matrix form

$$\begin{bmatrix} 4 & -3 \\ -3 & 9 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

This is in the form of ohm's law

$$[R][I] = [V]$$

$i_1$ and $i_2$ and be found by Cramer's rule :

$$i_1 = \frac{\begin{vmatrix} 5 & -3 \\ 0 & 9 \end{vmatrix}}{\begin{vmatrix} 4 & -3 \\ -3 & 9 \end{vmatrix}} = \frac{45}{36-9} = \frac{45}{27} = 1.667 \text{ Amp.}$$

Similarly $i_2$ can be determined :

$$i_2 = \frac{\begin{vmatrix} 4 & 5 \\ -3 & 0 \end{vmatrix}}{\begin{vmatrix} 4 & -3 \\ -3 & 9 \end{vmatrix}} = \frac{15}{36-9} = 0.555 \text{ Amp.}$$

**Example 3.8.** *Solve the above example 3.6 by Maxwell's node method.*

**Solution.** The circuit is redrawn in Fig. 3.38 and one node is taken as reference (zero volt) and the other is assigned the voltage $v$.

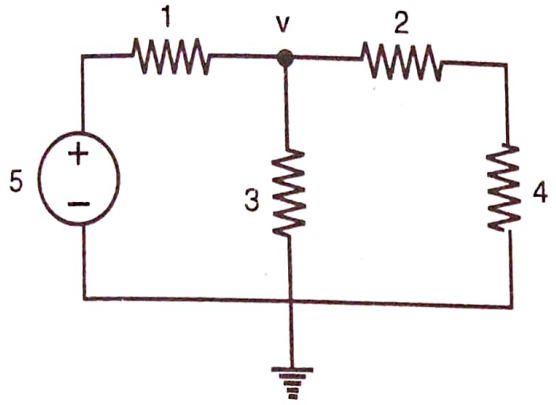



Fig. 3.38.

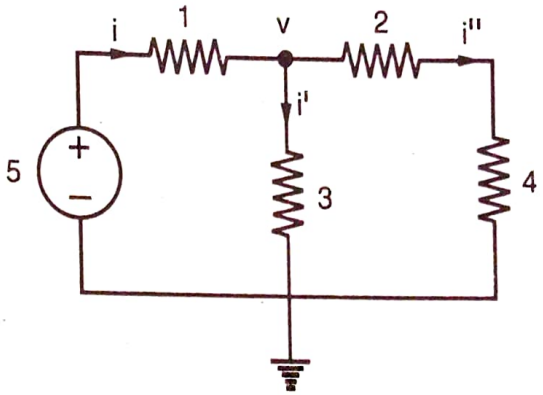



Fig. 3.39.

Writing the KCL equation at the node :

$$\frac{v-5}{1}+\frac{v}{3}+\frac{v}{6}=0$$

$$v\left[1+\frac{1}{3}+\frac{1}{6}\right]=5$$

$$\therefore \qquad v=\frac{30}{9}=3.333 \text{ volts.}$$

So the current passing through 1 Ω resistor $i=\frac{5-v}{1}=1.667$ Amp

Current passing through 3 Ω resistor $=i'=\frac{v}{3}=1.11$ Amp

and the current passing through 2 and 4 Ω resistors $=i''=\frac{v}{6}=0.555$ Amp.

**Example 3.9.** *Find i by Maxwell loop (mesh) method.*

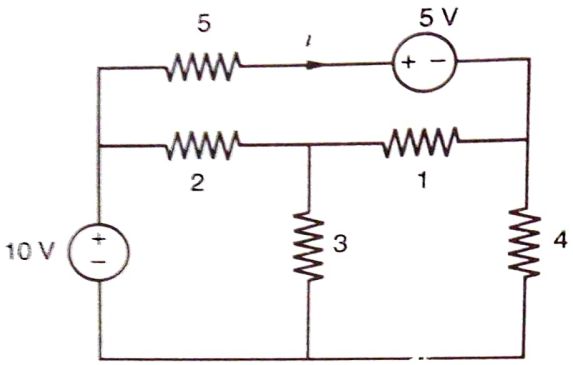



Fig. 3.40.

**Solution.** We select the loop as shown in Fig. 3.41.

We have to calculate $i_1$. So we apply Maxwell loop method (KVL) in all the three loops.

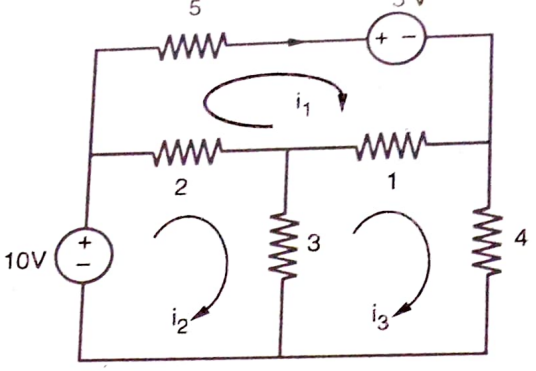



Fig. 3.41.

In loop 1 : $5\, i_1 + 1\,(i_1 - i_3) + 2\,(i_1 - i_2) = -5$

$\Rightarrow$ $8\, i_1 - 2\, i_2 - i_3 = -5$ ...(*i*)

In loop 2 : $2\,(i_2 - i_1) + 3\,(i_2 - i_3) = 10$

$\Rightarrow$ $-2\, i_1 + 5\, i_2 - 3\, i_3 = 10$ ...(*ii*)

In loops 3 : $1\,(i_3 - i_1) + 4\, i_3 + 3\,(i_3 - i_2)$

$\Rightarrow$ $-i_1 - 3\, i_2 + 8\, i_3 = 0$ ...(*iii*)

After solving Equations (*i*), (*ii*) and (*iii*) we can calculate all the three currents. But since we are interested in $i_1$ only so here we use Crammer's rule.

Equations (*i*), (*ii*) and (*iii*) can be written in matrix form

$$\begin{vmatrix} 8 & -2 & -1 \\ -2 & 5 & -3 \\ -1 & -3 & 8 \end{vmatrix} \begin{vmatrix} i_1 \\ i_2 \\ i_3 \end{vmatrix} = \begin{vmatrix} -5 \\ 10 \\ 0 \end{vmatrix}$$

By Cramer's rule,

$$i_1 = \frac{\begin{vmatrix} -5 & -2 & -1 \\ 10 & 5 & -3 \\ 0 & -3 & 8 \end{vmatrix}}{\begin{vmatrix} 8 & -2 & -1 \\ -2 & 5 & -3 \\ -1 & -3 & 8 \end{vmatrix}}$$

$$= \frac{-5\,(40-9) + 2\,(80) - 1\,(-30)}{8\,(40-9) + 2\,(-16-3) - 1\,(6+5)}$$

$$= \frac{-155 + 160 + 30}{248 - 38 - 11}$$

$$= \frac{35}{199} = 0.1758 \text{ Amp.}$$

current $i$ in circuit of Fig. 3.41 is 0.1758 Amp.

**ample 3.10.** *Find out the value of current in the branch BE of the network shown ig. 3.42.*

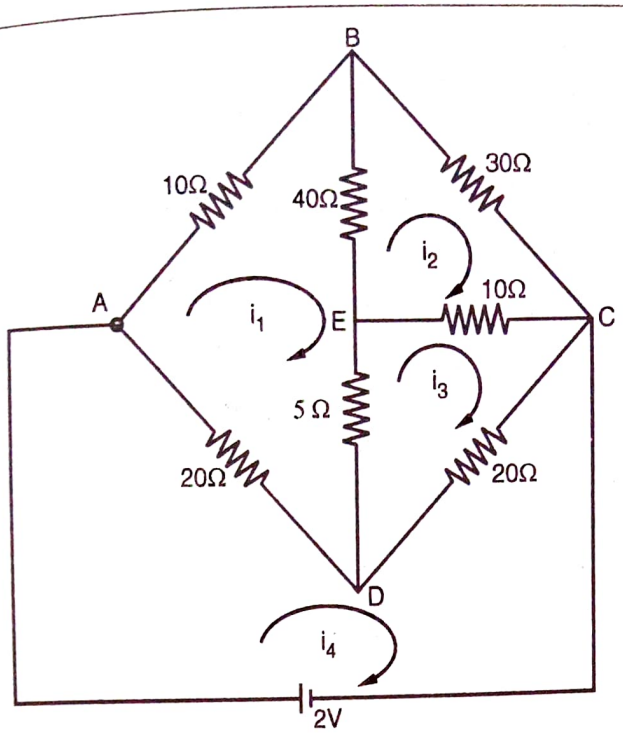



Fig. 3.42.

**Solution.** Using Mesh analysis we assume four mesh currents and write mesh equations.

Mesh equation for Mesh *ABEDA*,

$$75\, i_1 - 40\, i_2 - 5\, i_3 - 20\, i_4 = 0 \qquad ...(i)$$

Mesh equation for Mesh *BCEB*,

$$-40\, i_1 + 80\, i_2 - 10\, i_3 = 0 \qquad ...(ii)$$

Mesh equation for Mesh *ECDE*,

$$-5\, i_1 - 10\, i_2 + 35\, i_3 - 20\, i_4 = 0 \qquad ...(iii)$$

Mesh equation for Mesh *ADCA*,

$$-20\, i_1 - 20\, i_3 + 40\, i_4 - 2 = 0 \qquad ...(iv)$$

From equation (*iv*) we get $i_4$ and substitute it in equations (*i*), (*ii*) and (*iii*). We get,

$$i_4 = \frac{1}{40}(20\, i_1 + 20\, i_3 + 2)$$

$$-65\, i_1 + 50\, i_2 + 5\, i_3 + 1 = 0 \qquad ...(v)$$

$$40\, i_1 - 80\, i_2 + 10\, i_3 = 0 \qquad ...(vi)$$

$$15\, i_1 + 10\, i_2 - 25\, i_3 + 1 = 0 \qquad ...(vii)$$

By solving equations, (*v*), (*vi*) and (*vii*) we get,

$$i_1 = -0.048\,; \quad i_2 = -0.0345$$

$$i_3 = -0.0827\,; \quad i_4 = -0.01535$$

Current through branch $BE = i_1 - i_2 = -0.048 - (-0.0345) = -0.0135$

$= 0.0135$ Amp ( from $E$ to $B$).

**Example 3.11.** *Determine the current through the branch AB for the circuit shown in Fig. 3.43 using nodal analysis.* [RGPV, Bhopal June-July, 2006]

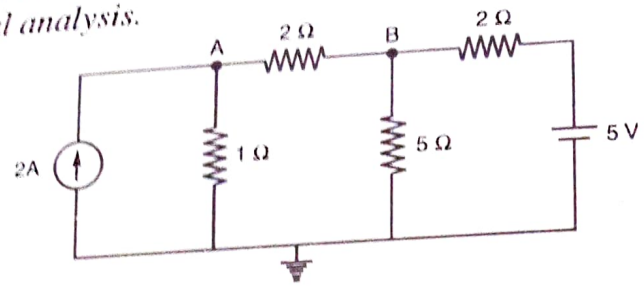



Fig. 3.43.

**Solution.** Node equation (KCL) at node node $A$

$$\frac{V_A}{1} + \frac{V_A - V_B}{2} = 2 \text{ A}$$

$$\Rightarrow \quad \frac{3}{2} V_A - \frac{V_B}{2} = 2 \qquad \ldots(i)$$

Node equation (KCL) at node $B$

$$\frac{V_B - V_A}{2} + \frac{V_B}{5} + \frac{V_B - 5}{2} = 0$$

$$-\frac{V_A}{2} + \frac{6}{5} V_B = \frac{5}{2} \qquad \ldots(ii)$$

From equation $(ii)$ $$V_B = \frac{5}{6}\left[\frac{5}{2} + \frac{V_A}{2}\right] \qquad \ldots(iii)$$

Put this value of $V_B$ in equation $(i)$, we get

$$V_A = 2.357 \text{ volt}$$

Put this value in equation $(iii)$, we get,

$$V_B = 3.065 \text{ volt}$$

Current through branch $AB = I_{AB} = \dfrac{V_A - V_B}{2} = \dfrac{2.357 - 3.065}{2}$

$$= -0.354 \text{ Amp.}$$

(Direction of current is from point $B$ to $A$)

**Example 3.12.** *In the given circuit. Find the current passing through 2 Ω resistor b node method. (Fig. 3.44.)*

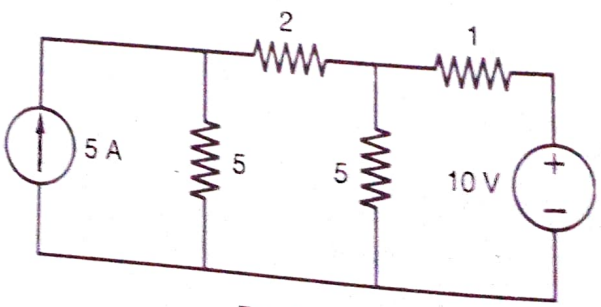



Fig. 3.44

**Solution.** There are two principle nodes and one reference (ground or datum) node. (Fig. 3.45).

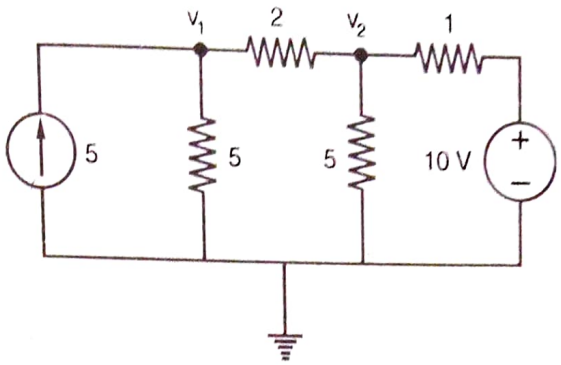



Fig. 3.45.

KCL at node 1 :
$$\frac{v_1}{5} + \frac{v_1 - v_2}{2} = 5 \qquad ...(i)$$

KCL at node 2 :
$$\frac{v_2 - v_1}{2} + \frac{v_2}{5} + \frac{v_2 - 10}{1} = 0 \qquad ...(ii)$$

Arrange these equations

$$\left(\frac{1}{5} + \frac{1}{2}\right) v_1 - \frac{1}{2} v_2 = 5 \qquad ...(iii)$$

$$-\frac{1}{2} v_1 + \left(\frac{1}{2} + \frac{1}{5} + 1\right) v_2 = 10 \qquad ...(iv)$$

From equation (*iii*)

$$0.7\, v_1 - 0.5\, v_2 = 5 \Rightarrow v_1 = \frac{5 + 0.5\, v_2}{0.7} \qquad ...(v)$$

From equation (*iv*)

$$-0.5\, v_1 + (1.7)\, v_2 = 10$$

Put value of $v_1$ from equation (*v*).

$$\Rightarrow \quad -0.5\left(\frac{5 + 0.5\, v_2}{0.7}\right) + 1.7\, v_2 = 10$$

$$\Rightarrow \quad -3.57 - 0.357\, v_2 + 1.7\, v_2 = 10$$

$$v_2 = 10.104 \text{ Volt.}$$

Put the value of $v_2$ in equation (*v*) :

$$v_1 = \frac{5 + 0.5\,(10.104)}{0.7} = 14.36 \text{ Volt.}$$

Now the current passing through 2 Ω resistor :

$$i = \frac{v_1 - v_2}{2} = \frac{14.36 - 10.104}{2} = 2.128 \text{ Amp.}$$

Now put the value of $i_2$ in equation (i) $i_1$ 17

**Example 3.17.** *Use Nodal analysis to determine the voltage across BC and the current in the 12 V source.* *(RGPV Bhopal, June 2007)*

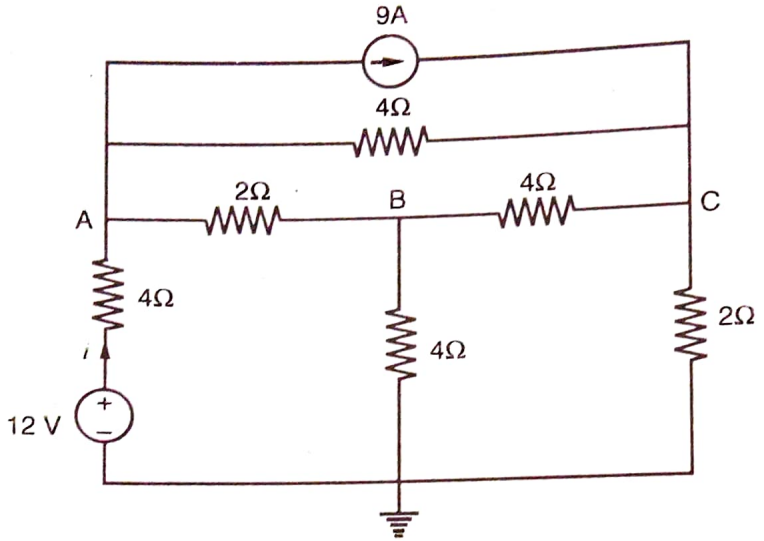



Fig. 3.52.

**Solution.** We apply Maxwell node method

KCL at node $A$ :

$$\frac{V_A - 12}{4} + \frac{V_A - V_B}{2} + \frac{V_A - V_C}{4} + 9 = 0$$

$$\left[\frac{1}{4} + \frac{1}{2} + \frac{1}{4}\right] V_A - \frac{1}{2} V_B - \frac{1}{4} V_C = -9 + 3$$

$$V_A - 0.5\, V_B - 0.25\, V_C = -6 \qquad \ldots(i)$$

KCL at node $B$ :

$$\frac{V_B - V_A}{2} + \frac{V_B}{4} + \frac{V_B - V_C}{4} = 0$$

$$-\frac{1}{2} V_A + \left[\frac{1}{2} + \frac{1}{4} + \frac{1}{4}\right] V_B - \frac{1}{4} V_C = 0$$

$$-0.5\, V_A + V_B - 0.25\, V_C = 0 \qquad \ldots(ii)$$

KCL at node $C$ :

$$\frac{V_C}{2} + \frac{V_C - V_B}{2} + \frac{V_C - V_A}{4} = 9$$

$$-0.25\, V_A - 0.25\, V_B + V_C = 9 \qquad \ldots(iii)$$

We solve this problem by Cramer's rule.

Write the equations $(i)$, $(ii)$ and $(iii)$ in matrix form

$$\begin{bmatrix} 1 & -0.5 & -0.25 \\ -0.5 & 1 & -0.25 \\ -0.25 & -0.25 & 1 \end{bmatrix} \begin{bmatrix} V_A \\ V_B \\ V_C \end{bmatrix} = \begin{bmatrix} -6 \\ 0 \\ 9 \end{bmatrix}$$

The determinant of matrix

$$\Delta = (1 - 0.0625) + 0.5\,(-0.5 - 0.0625) - 0.25\,(0.125 + 0.25)$$

$$= 0.9375 - 0.2812 - 0.09375 = 0.5625$$

By Cramer's rule :

$$V_A = \frac{\begin{vmatrix} -6 & -0.5 & -0.25 \\ 0 & 1 & -0.25 \\ 9 & -0.25 & 1 \end{vmatrix}}{0.5625} = \frac{-2.25}{0.5625} = -4 \text{ volt}$$

$$V_B = \frac{\begin{vmatrix} 1 & -6 & -0.25 \\ -0.5 & 0 & -0.25 \\ -0.25 & 9 & 1 \end{vmatrix}}{0.5625} = \frac{0}{0.5625} = 0 \text{ volt}$$

and

$$V_C = \frac{\begin{vmatrix} 1 & -0.5 & -6 \\ -0.5 & 1 & 0 \\ -0.25 & -0.25 & 9 \end{vmatrix}}{0.5625} = \frac{4.5}{0.5625} = 8 \text{ volt}$$

Voltage across branch $BC = V_{BC} = V_B - V_C$

$$= 0 - 8 = -8 \text{ volt}$$

Current in source $i = \frac{12 - V_A}{4} = \frac{12 - (-4)}{4} = 4$ Amp.

**Example 3.18** *Using nodal analysis, find the current through the 10 Ω resistor in Fig. 3.53.* (RGPV Bhopal, Dec. 2010)

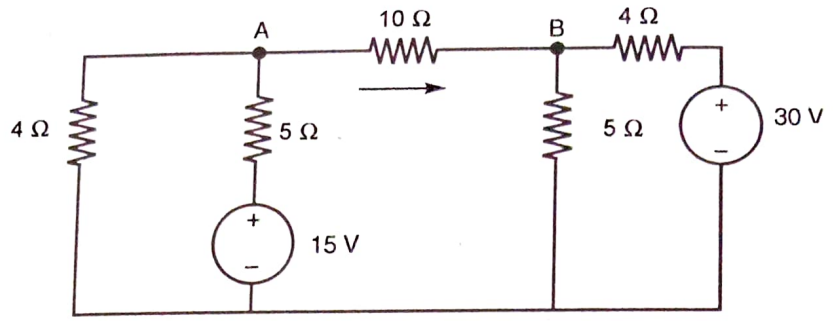



Fig. 3.53.

**Solution.** For finding the current passing through 10 Ω resistor, we have to calculate the voltages of nodes $A$ and $B$.

KCL at node $A$ :

$$\frac{V_A}{4}+\frac{V_A-15}{5}+\frac{V_A-V_B}{10}=0$$

$$V_A\left[\frac{1}{4}+\frac{1}{5}+\frac{1}{10}\right]-\frac{V_B}{10}=\frac{15}{5}$$

$$0.55\,V_A-0.1\,V_B=3 \quad \ldots(i)$$

KCL at node $B$ :

$$\frac{V_B-V_A}{10}+\frac{V_B}{5}+\frac{V_B-30}{4}=0$$

$$-\frac{V_A}{10}+\left[\frac{1}{10}+\frac{1}{5}+\frac{1}{4}\right]V_B=\frac{30}{4}$$

$$-0.1\,V_A+0.55\,V_B=7.5 \quad \ldots(ii)$$

From equation $(i)$

$$V_A=\frac{3+0.1\,V_B}{0.55}$$

Put this value of $V_A$ in equation $(ii)$

$$-0.1\left[\frac{3+0.1\,V_B}{0.55}\right]+0.55\,V_B=7.5$$

$$-0.5454-0.01818\,V_B+0.55\,V_B=7.5$$

$$-0.5454+0.5318\,V_B=7.5$$

$$V_B=15.1286 \text{ volts}$$

Now,

$$V_A=\frac{3+0.1\,(15.1286)}{0.55}=8.2052 \text{ Volts}$$

Current passing through 10 Ω resistor in the direction shown in Fig. 3.53.

$$I=\frac{V_A-V_B}{10}=-0.6923 \text{ Amp.}$$

The direction of current is from node $B$ to $A$.